Electrical Orientation

Course Objectives:

- Familiarize you with basic electrical terms and symbols
- Familiarize you with basic AC and DC electrical theory
- Familiarize you with transformers
- Familiarize you with electric motors
- Familiarize you with the use of basic electrical test equipment
- Familiarize you with basic electrical troubleshooting techniques
- Familiarize you with NFPA 70E and SGOP-0022.

Basic Electrical Terminology

Voltage: The driving force which causes electrical current to flow. Measured in volts and sometimes referred to as Electromotive force.

Ohm’s Law:

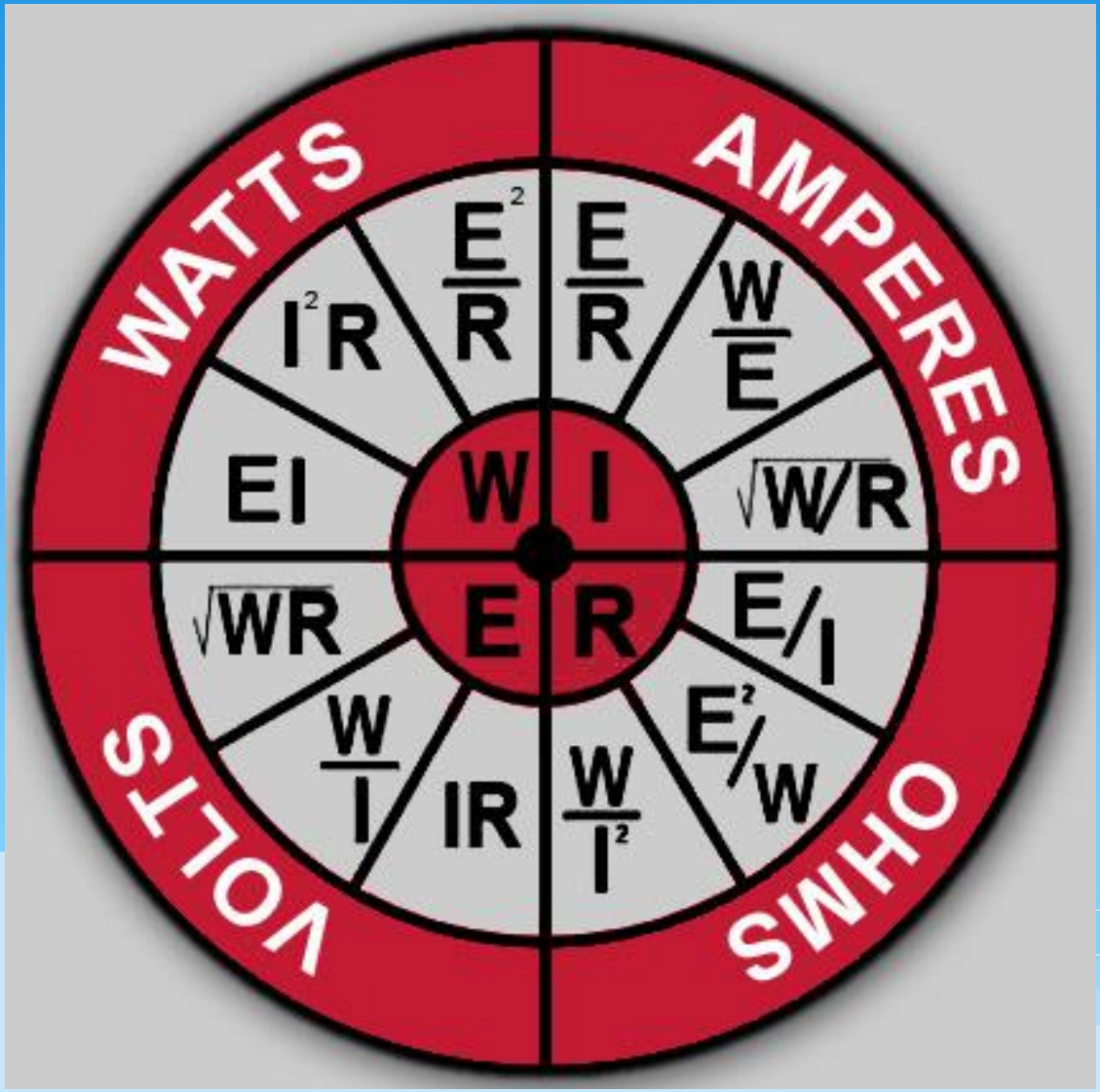
In any electrical circuit;

The voltage is directly proportional to current and/or resistance.

Mathematically stated: $E = I \times R$

(E=voltage, I=current, R=resistance)

Ohm's Law Wheel



- **Conductor**

  A substance, usually metal which allows electrical current to flow freely.

- **Insulator**

  A substance, usually plastic or rubber which does **not** allow electrical current to flow freely.

- **Resistance**

  The property of a circuit element which opposes the flow of electrical current. Measured in Ohms.

- **Resistor**

  A device which opposes the flow of electrical current.

- **Capacitance**

  The property of a circuit which opposes a change in voltage. Measured in Farads.

- **Capacitor**

  A device which stores electrical energy in an electrostatic field.

  A capacitor will conduct AC but block DC.

- **Ammeter**

  A device used to measure the flow of electrical current.

- **Clamp on ammeter**

  A device used to measure the flow of electrical current without the need to be connected in series with the circuit.

- **Ohmmeter**

  A device used to measure resistance.

- **Megger**

  A device used to measure insulation leakage or integrity.

- **Series circuit**

  A circuit in which all circuit components are connected so that there is **one and only one** path for current flow

- **Parallel circuit**:

  A circuit in which circuit components are connected so that there is **two or more** paths for current flow.

- **Open Circuit**

  A lack of continuity in a circuit, such as a broken wire.

- **Short Circuit**

  A direct connection from one power line to another. This results in a fault current which will normally cause a fuse to blow or circuit breaker to trip.

- **Ground fault**

  A connection from a power line to ground (earth, metal cabinets, etc.). Frequently caused by water in electrical equipment.

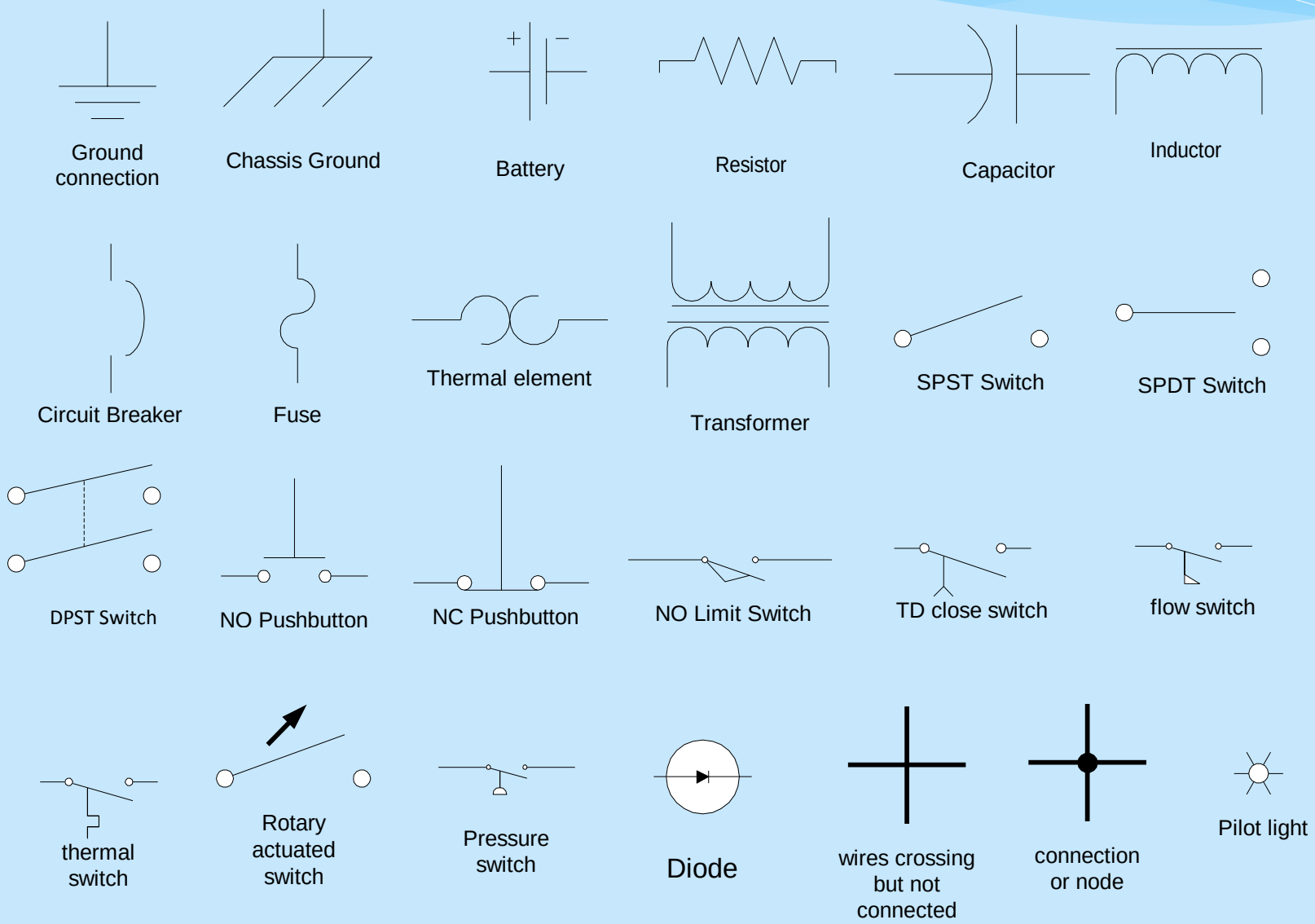
- **Bonding**

  Intentionally connecting exposed metal chassis parts, electrical conduit, etc.. together and to ground (earth).

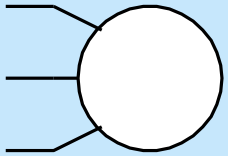
- **Ground Fault Circuit Interrupter (GFCI)**

  A safety device used to interrupt power to equipment with a ground fault.

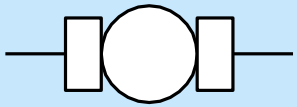
Electrical Symbols



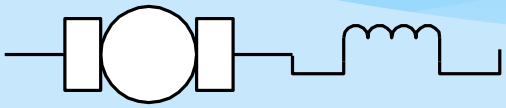
Electrical Symbols



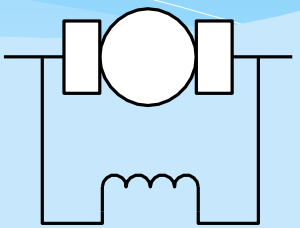
3 phase motor



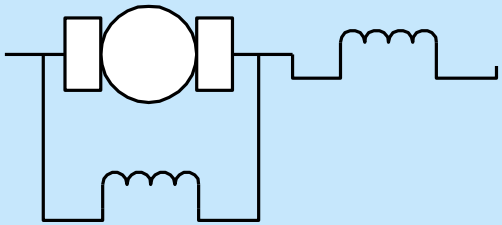
Universal motor



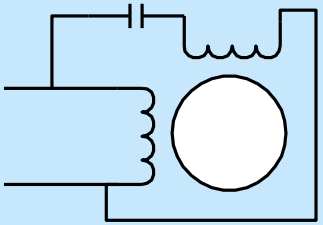
Series DC Motor



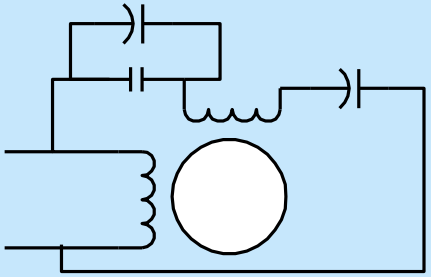
Shunt DC Motor



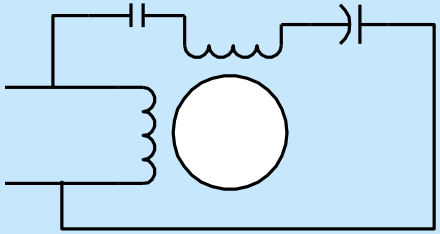
Compound DC motor



Split phase
induction motor



Capacitor start / capacitor run
induction motor



Capacitor start induction motor

Basic Principles

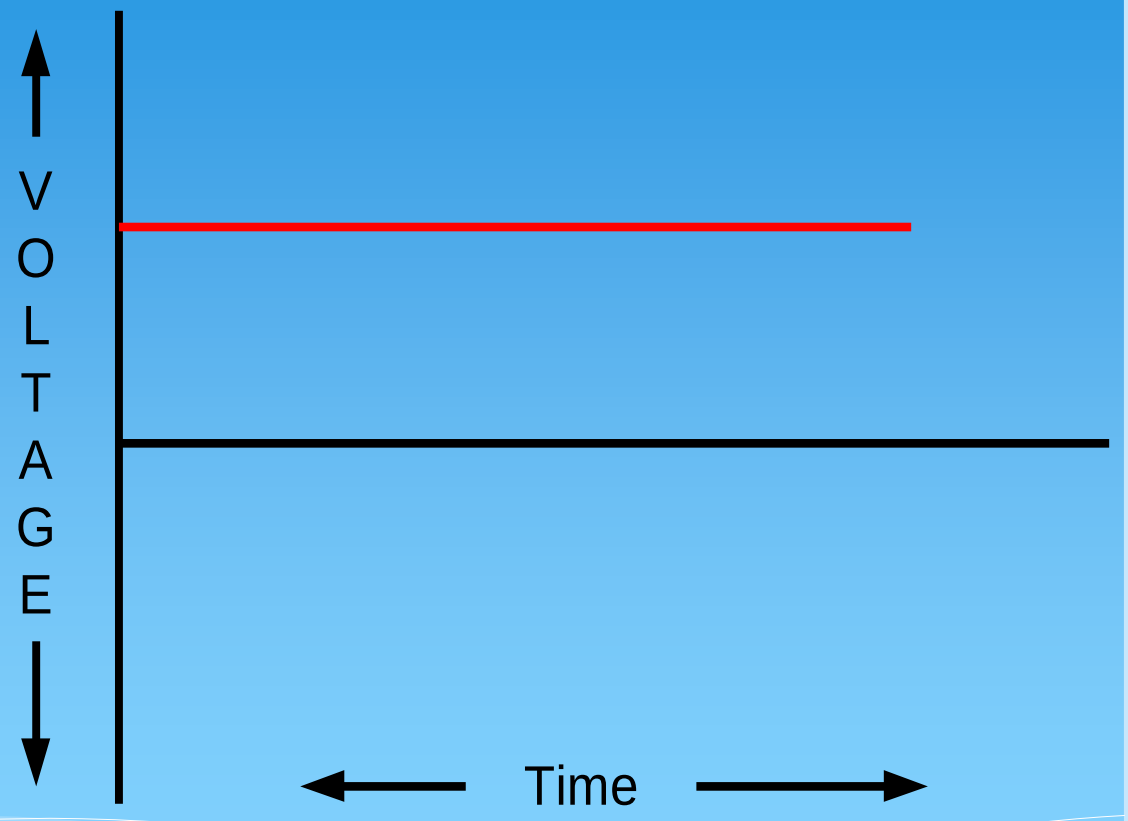
- There must be a **complete** electrical path for current to flow.

- In a given circuit, if voltage is held constant and resistance decreases, the current will increase (Ohm’s Law).

DC Power

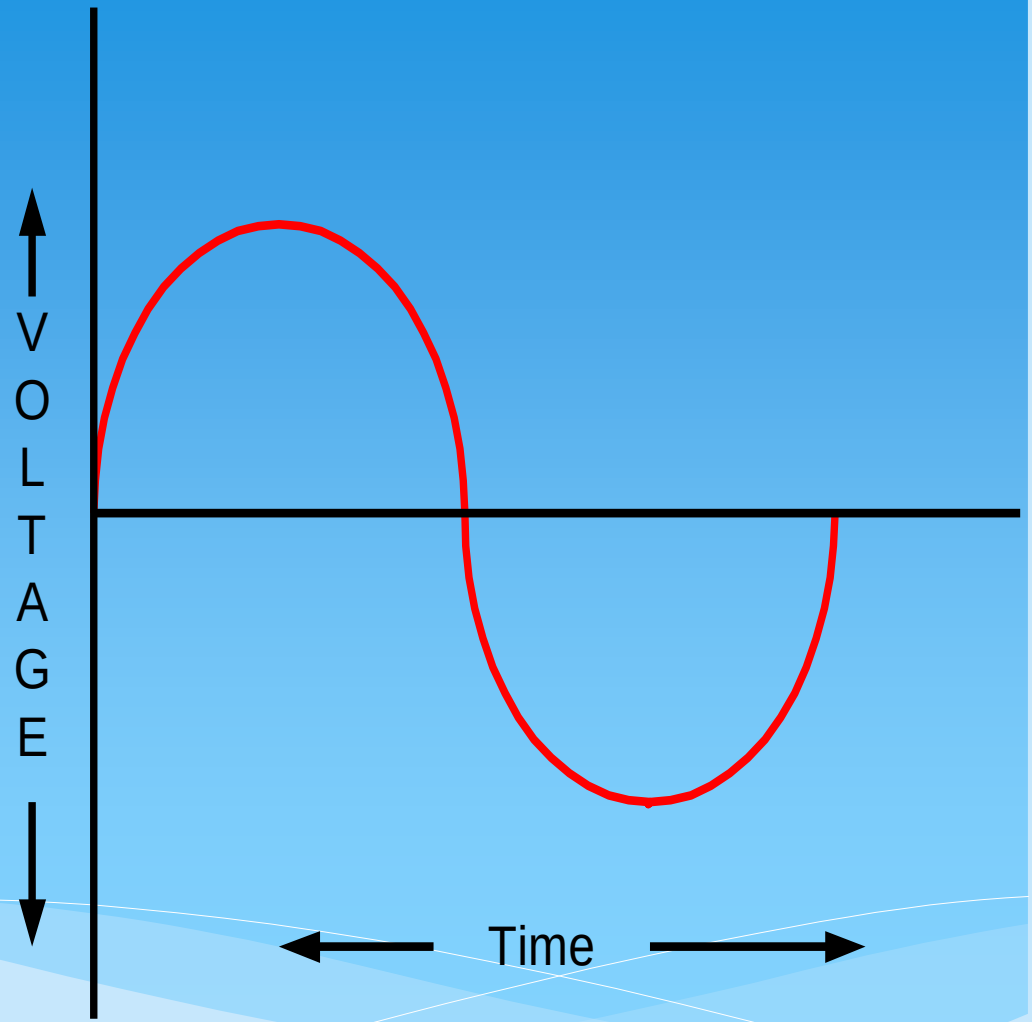
Voltage is either positive or negative

Polarity does not change over time



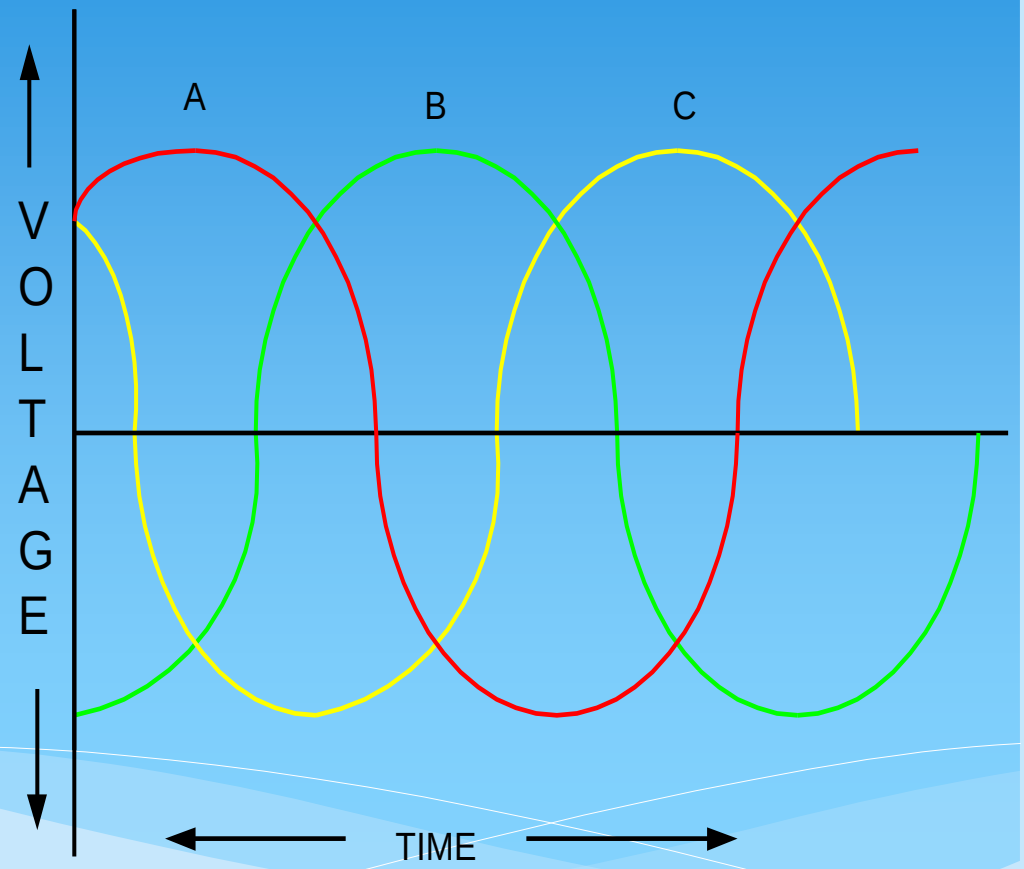
AC Power

Voltage and polarity change over time



3 Phase AC Power

3 separate AC voltages,120 electrical degrees apart

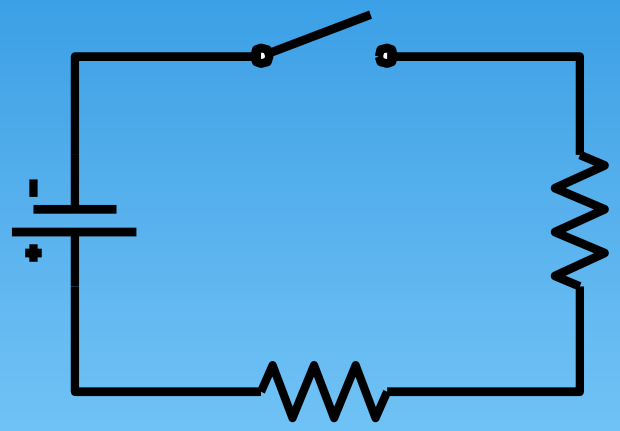


Series Circuit

The current is common to all circuit elements.

The voltage developed by each circuit element may be different.

Resistance is additive (i.e. Any additional resistance added in a series circuit will increase the total circuit resistance).



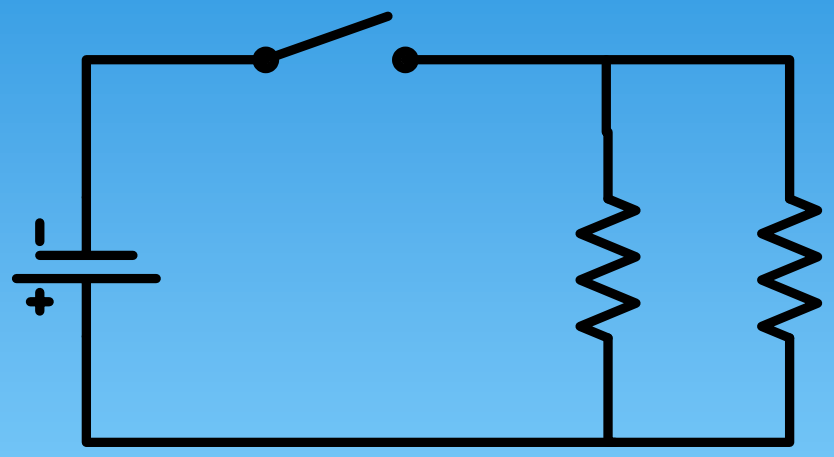
Series Circuit

Parallel Circuit

The voltage is common to all parallel branches of the circuit.

The current through each parallel branch may be different.

Resistance is not additive (i.e. Each parallel resistance added to the circuit will decrease the total circuit resistance.



Parallel circuit

Transformers

- Consists of primary and secondary windings wound on a laminated steel alloy core
- Transformers are Capable of lowering or raising voltage depending on turns ratio of primary to secondary windings
- Transformers can also be used to stabilize voltages as well, these types are called isolation transformers.
- Primary connections are typically labeled “H”
- Secondary connections are typically labeled “X”

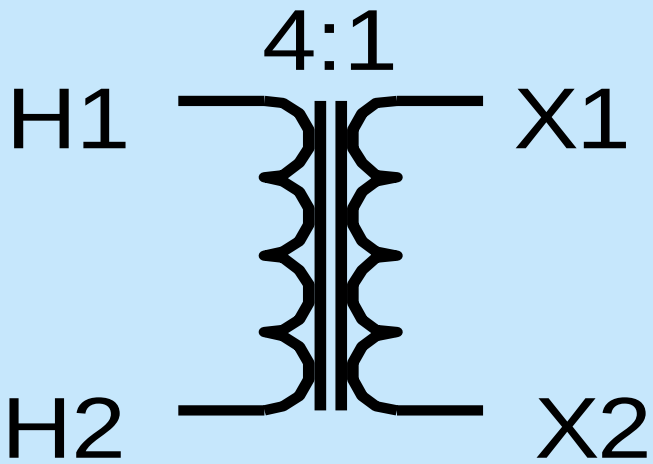
Transformers

- Transformers can be cooled using various methods, including oil, water, and air.
- A good example of a transformers' use is when high voltage power lines are converted to a lower voltage, **120/240** volts, before it enters your home.

Step Down Transformer

- Voltage change from primary to secondary is proportional to the turns ratio
- Example: 480V Primary to 120V Secondary

480V

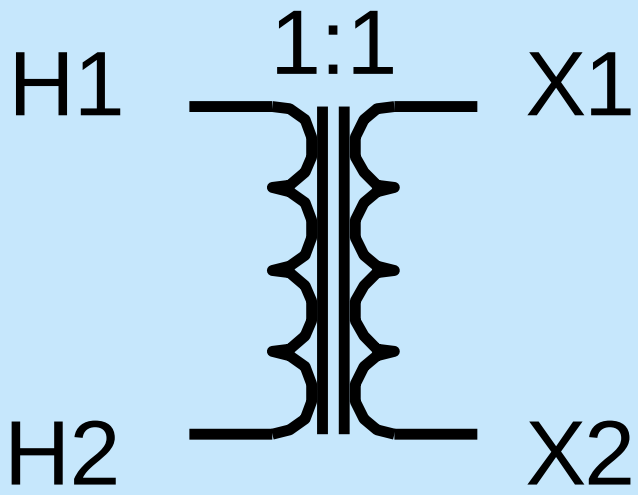


120V

Step down transformer

Isolation Transformer

- Used for the isolation of one circuit from another. This will isolate grounds and noise from one AC circuit to another.



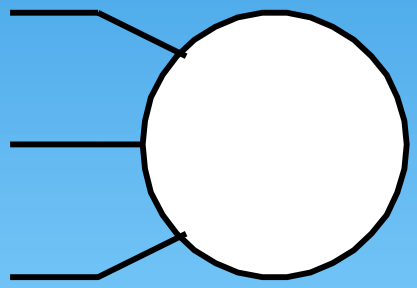
Isolation transformer

Types of Electric Motors

- 3 Phase induction motors
- Single phase induction motors
- Universal motors
- DC motors

3 Phase Motors

The rotating direction of a 3 phase motor can be reversed by swapping **any 2** motor leads either at the load side of the motor starter or disconnect.



3 phase motor

Test Equipment

Multimeters are used for:

- Voltage measurement
- Current measurement
- Resistance measurement

Clamp-on Ammeter

- The clamp-on ammeter is used to measure current without the need to break the circuit, the meter must be rated for the type and amount of current and voltage.

  Note: This is primarily used on AC current but can also be used to measure high current DC.

- The proper manner to use a clamp-on ammeter is:

  Open the spring loaded clamp, inserting one wire at a time in the loop, close the clamp completely and take the measurement. For 3 phase, each leg should be measured separately.

Megohmmeter (Megger)

- A “megger” is used to measure insulation resistance by inserting a high voltage across the insulation. It’s primary purpose is to check the integrity of electric motor windings and/or conductor insulation.
- The motor and/or wiring being tested must be de-energized and disconnected from the source.
- The megger is connected between the motor leads and the motor frame, or ground.

Megger

Because the Megger works by introducing a voltage *into* a circuit, it is recommended that only **“Qualified Persons”**, properly trained and authorized to use this meter utilize this tool during start-up testing and troubleshooting.

Voltage Measurement

1) Ensure all safety requirements are met prior to attempting voltage measurement on a live circuit.

2) Ensure that multimeter is set to the proper voltage type (AC or DC) and voltage range for the expected voltage.

   Note: Many multimeters automatically switch to the proper voltage range.

3) For DC voltage, observe polarity of voltage. Connect the red test lead to the positive polarity and the black test lead to the negative polarity.

4) For voltage measurement, the multimeter leads must be connected in shunt (parallel) with the item under test.

Current Measurement (Series)

1) In order to measure current with a multimeter, the meter must be connected in series with the circuit under test. To accomplish this, first de-energize the circuit and connect the meter leads in series with the circuit.

2) Ensure that multimeter is set to the proper current type (AC or DC) and current range for the expected maximum current.

   Note: Many multimeters automatically switch to the proper current range. You may also have to connect the meter leads to different jacks on the multimeter to measure current.

3) Ensure that the multimeter is properly rated to carry the maximum anticipated circuit current, or meter damage could result.

4) After connecting the meter in series with the circuit, and ensuring that it is configured correctly, energize the circuit and observe the reading.

5) You must de-energize the circuit , remove the meter leads and restore the original circuit connections after you finish measuring the current.

Resistance measurement

1) **Ensure that the circuit to be tested is de-energized**.
2) You may have to isolate the circuit element you want to test, because if there are any parallel circuits, they will give you a misleading meter reading.
3) Ensure that the meter is set to the proper resistance range for the measurement you are taking.
   Note many multimeters automatically switch to the proper resistance range.
4) The multimeter leads are placed across the device under test.

“Qualified Person” - Definition

Qualified Person:

One who has skills and knowledge related to the construction and operation of the electrical equipment and installation and ***has received safety training to recognize and avoid the hazards involved***. These individuals have been trained and are competent at avoiding the hazards of working on or near electrically energized components and are authorized by management to perform such work.

Examples of hazards include:

1. Contact with potentially energized surfaces (50 to 600 volts to ground) (Shock)
2. Arc flash due to short circuit or component failures.
3. The Arc blast/thermal energy that may cause facial burns, clothing fires and possible blinding if a failure occurs.

NOTE: At JJV, there are two classifications of Qualified Persons:

Green Card and Blue Card.

The SGOP-0022 documents the certification process and the type of work allowed for each of the classifications.

Establishing an Electrically Safe Work Condition

1. Apply Lock and Tag in accordance with the applicable HECP (refer to the current revision of SGOP-0028) Use the Left (Off) Hand Rule when operating disconnects or breakers.
2. Use a TEGAM or FLUKE T5 or Metrix MX 57EX (intrinsically safe areas only) Tester.
3. Check the meter is operating correctly on a known voltage source.
4. Verify the circuit de-energized by checking each conductor both phase-to- phase and phase-to ground.
5. Re-Check the meter is operating correctly on a known voltage source.
6. Electrically Safe Work condition exists only if all of the above are followed and the line side of the main disconnecting device is Finger Touch Protected such that the risk of accidental contact is eliminated and there are no other energized components present. No electrical PPE is required and conductive tools are permitted to be used.
7. The Electrically Safe Work condition shall be re-established if power is applied.
8. The circuit shall be considered energized until it is placed in an Electrically Safe Work Condition. Use of control devices or interlock circuits is not an acceptable method of lockout/tagout.

The equipment is now in an electrically safe work condition.

Meters for Establishing an Electrically Safe Work Area

Job Hazard Analysis

A Job Hazards Analysis shall be performed for each activity that has the potential for personnel to be exposed to injury due to electrical hazards. (From SGOP-0022)

- Isolation means (fed from)
- Determine ARC Flash Boundary
- Determine ARC Flash Energy level at the working distance (18 or 24 inches)
- Determine and don proper PPE (Arc Flash and shock protection)
- Determine nominal voltage
- Determine Shock Boundaries
- Limited Approach Boundary
- Restricted Approach Boundary
- Review for Special Operating Conditions
- Description of the safe work practices to be used
- Station Attendant (Brief as required)
- Other means to restrict access (as needed)
- Insulated/Voltage Rated tools and test equipment as required
- Non-conductive lighting (as needed)
- Apply adequately rated personal protective ground sets (as needed)
- Energized Electrical Work Permit (Form Control No. VCF-2335) (as needed)

Job Hazard Analysis

- The results of the Job Hazards Analysis shall be communicated to all persons involved with the work activity and with any Un-Qualified persons that could be exposed to the hazards.
- This is required as a pre-job task and is not required to be formally documented.
- Employees are not allowed to reach blindly into areas that might contain exposed energized electrical conductors or circuit parts where electrical hazards exists.

SPECIAL OPERATING CONDITIONS

- Special Operating Conditions are those areas, installation methods, equipment, work practices, etc. unique to the location that have the potential to present safety hazards to personnel, that would not normally be encountered at other similar facilities.

- Special operating conditions include Electrically classified areas including the Isopropyl Alcohol and chemical storage areas, wet locations (sterilizers), troubleshooting IS barriers, waste plastic/resin areas, and areas with special lighting.

- Where flammable materials are present, only explosion-proof or intrinsically safe electrical equipment, including electrical test equipment, shall be used that is in compliance with the appropriate codes and regulations regarding classified areas.

PPE
AF PPE 0 Requirements

- Safety glasses or goggles.
- Hearing protection (ear canal inserts)
- Arc rated long sleeve shirt or a lab coat (capable of fastening at the neck) with a minimum arc rating of 4 cal/cm2
- Untreated Natural Fiber Pants (long)
- Heavy duty leather gloves (as needed)

**(SEE current FMJSA – 0038 for specifics)*

PPE

- Personal Protective Equipment:

  Consult **Form Control No.VWA-0836,** for its selection, care, handling, maintenance, inspection, and use of PPE requirements. **Form Control No.VWA-0834** contains the purchasing specifications of PPE and Other Electrical Protective Equipment. Refer to the arc flash label to identify personal protective equipment needed for protection against electrical hazards.
- For equipment that does not have an Arc Flash Label, refer to **Form Control No.VWA-0839**

FMJSA – 0038 - Entering a Category 0 Electrical Cabinet

FMJSA – 0039 - Entering a Category 1 Electrical Cabinet

FMJSA – 0040 - Entering a Category 2 Electrical Cabinet

FMJSA – 0041 - Entering a Category 3 Electrical Cabinet

FMJSA – 0042 - Entering a Category 4 Electrical Cabinet

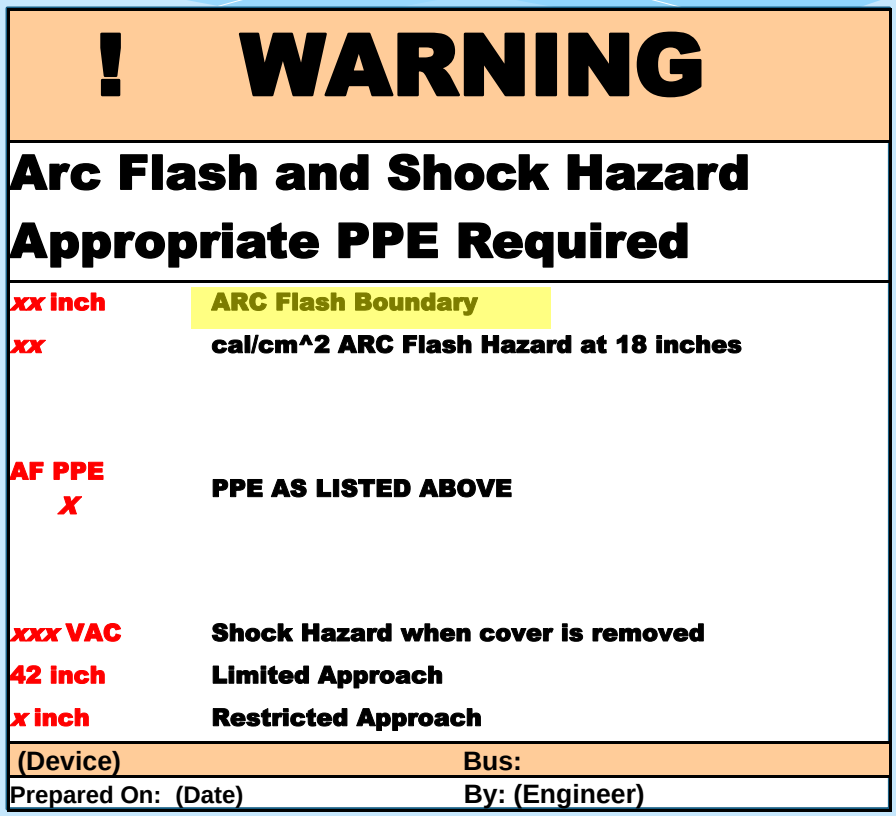
Purpose of Arc Flash Warning Labels

- Identifies Arc Flash Hazard Boundary
- Minimum PPE requirements
- Hazard Category
- Voltage levels
- Limited and Restricted Approach Boundaries

! WARNING	
Arc Flash and Shock Hazard Appropriate PPE Required	
xx inch	ARC Flash Boundary
xx	cal/cm^2 ARC Flash Hazard at 18 inches
AF PPE *X*	PPE AS LISTED ABOVE
xxx VAC	Shock Hazard when cover is removed
42 inch	Limited Approach
x inch	Restricted Approach
(Device)	Bus:
Prepared On: (Date)	By: (Engineer)

ARC Flash Label Review

- Arc Flash Boundary – PPE is REQUIRED
  - Calculation
  - Available Current – transformer size
  - Clearing Time – Fuse/Breaker
- Inside this distance energy of 5 J/cm^2 (1.2 cal/cm^2) can be generated
- Non-curable - 2nd Degree Burn



! WARNING

Arc Flash and Shock Hazard
Appropriate PPE Required

xx inch	ARC Flash Boundary
xx	cal/cm^2 ARC Flash Hazard at 18 inches
AF PPE *X*	PPE AS LISTED ABOVE
xxx VAC	Shock Hazard when cover is removed
42 inch	Limited Approach
x inch	Restricted Approach

(Device) Bus:

Prepared On: (Date) By: (Engineer)

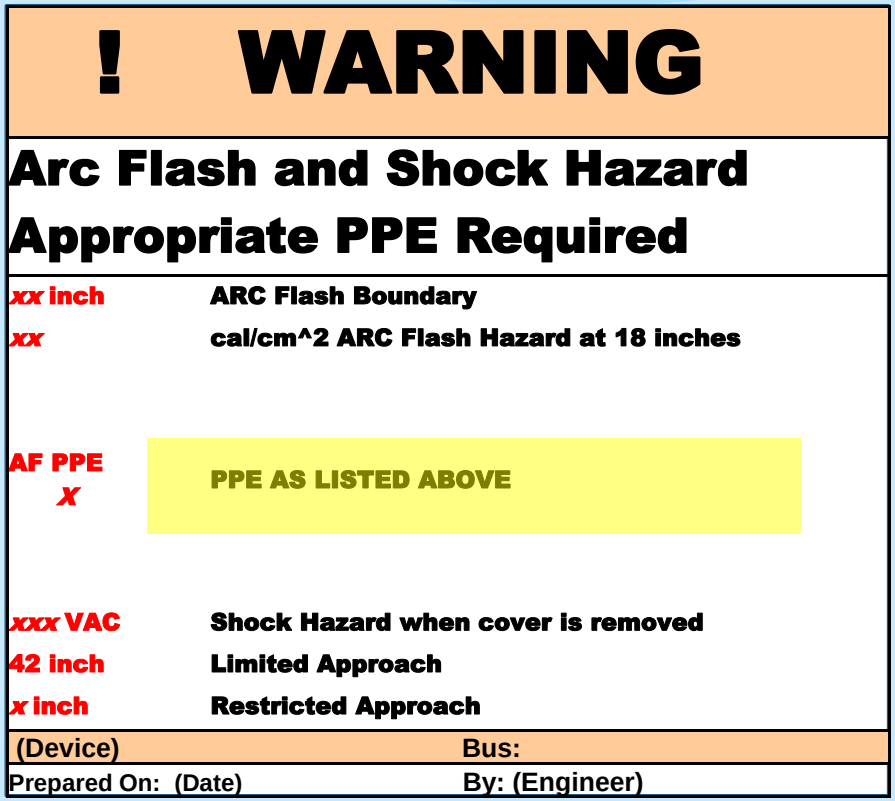
ARC Flash Label Review

* 18 Inch calculation
* Approximate working distance from live electrical
* Basis for PPE
* PPE – NFPA 70E (12) Table 130.7(C)(16)

! WARNING	
Arc Flash and Shock Hazard Appropriate PPE Required	
xx inch	ARC Flash Boundary
xx	cal/cm^2 ARC Flash Hazard at 18 inches
AF PPE *X*	PPE AS LISTED ABOVE
xxx VAC	Shock Hazard when cover is removed
42 inch	Limited Approach
x inch	Restricted Approach
(Device)	Bus:
Prepared On: (Date)	By: (Engineer)

ARC Flash Label Review

* AF PPE – Required PPE Level
* PPE rated greater than the expected energy level at 18”
* Lab coats are required in most labs
* FMJSA for all Classes of PPE
* VWA – 0839 PPE reference



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AF PPE *X*	PPE AS LISTED ABOVE
xxx VAC	Shock Hazard when cover is removed
42 inch	Limited Approach
x inch	Restricted Approach
(Device)	Bus:
Prepared On: (Date)	By: (Engineer)

ARC Flash Label Review

* Highest Rated voltage in the cabinet
* Determines Approach boundaries
* Approach boundaries based on Shock protection
* PPE required inside all
* Insulated tools and meters inside all

! WARNING	
Arc Flash and Shock Hazard Appropriate PPE Required	
xx inch	ARC Flash Boundary
xx	cal/cm^2 ARC Flash Hazard at 18 inches
AF PPE *X*	PPE AS LISTED ABOVE
xxx VAC	Shock Hazard when cover is removed
42 inch	Limited Approach
x inch	Restricted Approach
(Device)	Bus:
Prepared On: (Date)	By: (Engineer)

ARC Flash Label Review

- Limited Approach – All must wear PPE
- Remains until electrically safe work area is established
- Unqualified can cross if escorted by a Blue Badge, wearing PPE, & informed of hazards.
- Insulated tools and meters required

! WARNING	
Arc Flash and Shock Hazard Appropriate PPE Required	
xx inch	ARC Flash Boundary
xx	cal/cm^2 ARC Flash Hazard at 18 inches
AF PPE *X*	PPE AS LISTED ABOVE
xxx VAC	Shock Hazard when cover is removed
42 inch	Limited Approach
x inch	Restricted Approach
(Device)	Bus:
Prepared On: (Date)	By: (Engineer)

ARC Flash Label Review

* Restricted Approach – PPE Required
* NO UNQUALIFIED
* Insulated tools and meters required

! WARNING	
Arc Flash and Shock Hazard Appropriate PPE Required	
xx inch	ARC Flash Boundary
xx	cal/cm^2 ARC Flash Hazard at 18 inches
AF PPE *X*	PPE AS LISTED ABOVE
xxx VAC	Shock Hazard when cover is removed
42 inch	Limited Approach
x inch	Restricted Approach
(Device)	Bus:
Prepared On: (Date)	By: (Engineer)

SHOCK HAZARD BOUNDRIES

Table 130.4(D)(a) Approach Boundaries to Energized Electrical Conductors or Circuit Parts for Shock Protection for Alternating-Current Systems (All dimensions are distance from energized electrical conductor or circuit part to employee.)

(1)	(2)	(3)	(4)
	Limited Approach Boundary[b]		Restricted Approach Boundary[b]; Includes Inadvertent Movement Adder
Nominal System Voltage Range, Phase to Phase[a]	Exposed Movable Conductor[c]	Exposed Fixed Circuit Part	
<50 V	Not specified	Not specified	Not specified
50 V–150 V[d]	3.0 m (10 ft 0 in.)	1.0 m (3 ft 6 in.)	Avoid contact
151 V–750 V	3.0 m (10 ft 0 in.)	1.0 m (3 ft 6 in.)	0.3 m (1 ft 0 in.)
751 V–15 kV	3.0 m (10 ft 0 in.)	1.5 m (5 ft 0 in.)	0.7 m (2 ft 2 in.)
15.1 kV–36 kV	3.0 m (10 ft 0 in.)	1.8 m (6 ft 0 in.)	0.8 m (2 ft 7 in.)
36.1 kV–46 kV	3.0 m (10 ft 0 in.)	2.5 m (8 ft 0 in.)	0.8 m (2 ft 9 in.)
46.1 kV–72.5 kV	3.0 m (10 ft 0 in.)	2.5 m (8 ft 0 in.)	1.0 m (3 ft 3 in.)
72.6 kV–121 kV	3.3 m (10 ft 8 in.)	2.5 m (8 ft 0 in.)	1.0 m (3 ft 4 in.)
138 kV–145 kV	3.4 m (11 ft 0 in.)	3.0 m (10 ft 0 in.)	1.2 m (3 ft 10 in.)
161 kV–169 kV	3.6 m (11 ft 8 in.)	3.6 m (11 ft 8 in.)	1.3 m (4 ft 3 in.)
230 kV–242 kV	4.0 m (13 ft 0 in.)	4.0 m (13 ft 0 in.)	1.7 m (5 ft 8 in.)
345 kV–362 kV	4.7 m (15 ft 4 in.)	4.7 m (15 ft 4 in.)	2.8 m (9 ft 2 in.)
500 kV–550 kV	5.8 m (19 ft 0 in.)	5.8 m (19 ft 0 in.)	3.6 m (11 ft 10 in.)
765 kV–800 kV	7.2 m (23 ft 9 in.)	7.2 m (23 ft 9 in.)	4.9 m (15 ft 11 in.)

Note (1): For arc flash boundary, see 130.5(A).

Note (2): All dimensions are distance from exposed energized electrical conductors or circuit part to employee.

[a] For single-phase systems above 250V, select the range that is equal to the system's maximum phase-to-ground voltage multiplied by 1.732.

[b] See definition in Article 100 and text in 130.4(D)(2) and Informative Annex C for elaboration.

[c] *Exposed movable conductors* describes a condition in which the distance between the conductor and a person is not under the control of the person. The term is normally applied to overhead line conductors supported by poles.

[d] This includes circuits where the exposure does not exceed 120V.

Approach Boundaries

Arc Flash Boundary-

Persons should not cross unless wearing PPE and under supervision of a Qualified person **(note: there is no correlation between arc flash and shock boundaries, not always farther than shock boundary)**

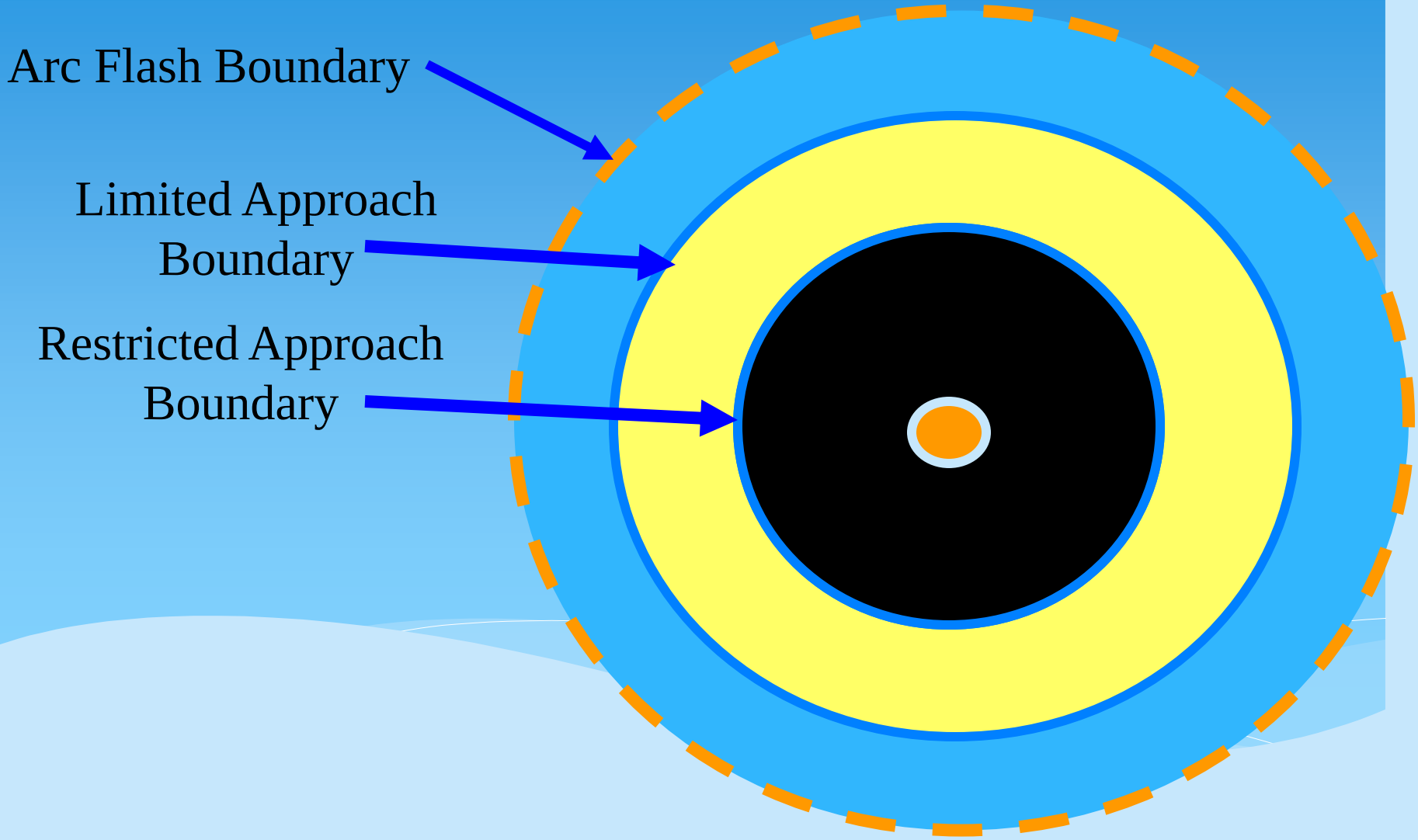
Limited Approach Boundary-

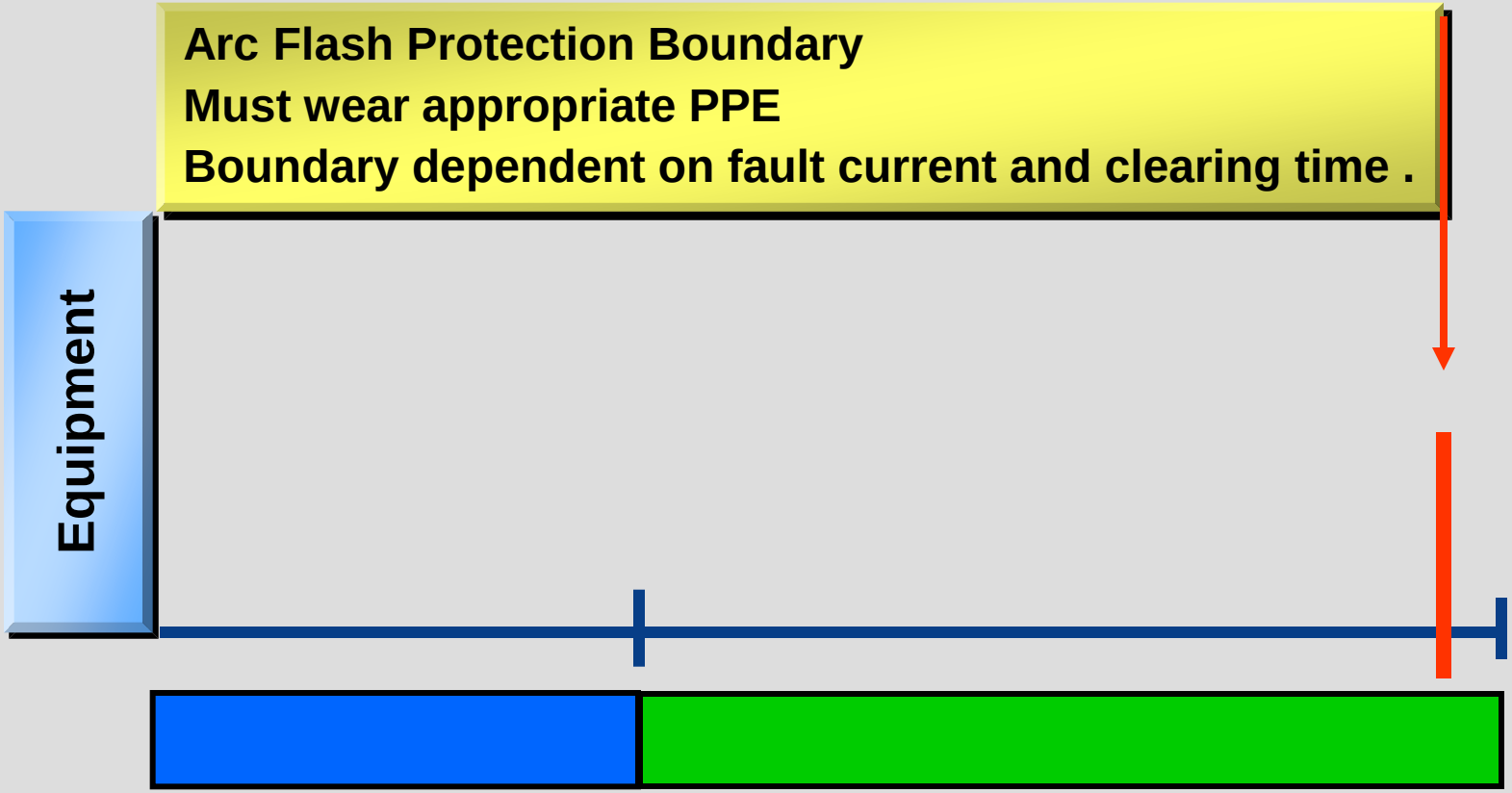
A Shock Hazard Exist. PPE is required in this area. Where there is a need for an unqualified person to cross for on the job training, a qualified person shall advise him or her of the possible hazards and continuously escort the unqualified person(s) while inside this Boundary

Restricted Approach Boundary-

A Shock Hazard Exist. PPE is required in this area. Only qualified individuals may be in this area.

Approach Boundaries





Restricted Shock Boundary: Qualified Persons Only

Limited Shock Boundary: Qualified or Unqualified Persons*

* Only if accompanied by Qualified Person

Note: shock boundaries dependent on system voltage level

PORTABLE ELECTRICAL CORD & PLUG EQUIPMENT

- Reference **Form Control No.VWA-0837** for proper use, care, maintenance, testing and purchasing specifications for cord and plug equipment.
- Reference **Form Control No. VWA-0833** for purchasing specifications for electrical plug and cord connected equipment and tools.

NOTE: Plugs and cords are prohibited as a substitute for permanent wiring.

PORTABLE ELECTRICAL CORD & PLUG EQUIPMENT

- **Ground fault circuit interrupter (GFCI):**

  GFCIs shall be utilized whenever an extension cord is used to operate a portable electric tool. The GFCI device shall be plugged directly into the electrical outlet followed by the extension cord and then by the portable electric tool. GFCI should be tested per the manufactures recommendations.

- **Extension Cords:**

  All flexible extension cords shall contain a grounding conductor. No more than 100 feet of extension cord, supplying rated current for the wire size - minimum 14-gauge (as marked on the cord by the manufacturer) shall be used on site. GFCI devices shall be used with all extension cords in industrial environments. GFCI devices shall be used in conductive areas. GFCI Pigtails must be rated the same as the extension cord it is plugged in to.

PORTABLE ELECTRICAL AND ELECTRONIC TEST EQUIPMENT

- The requirements for the use, care, inspection, maintenance is established in **Form Control No.VWA-0838** and purchasing specifications of test instruments and equipment is established in **Form Control No.VWA-0835**

- Portable electrical and electronic test equipment shall be periodically inspected and tested by a **Qualified Person** at least annually utilizing **Form Control No.VCF-2336.**

- Use – Only **Qualified Persons** shall perform testing work on electric circuits or equipment that is rated at 50 volts or more to ground.

Green Card

Green Card classification: Electrical Work Allowed

- Troubleshoot, take meter readings, inspect, and make adjustments to equipment settings when equipment is finger touch protected equipment and is **600 Volts or less**.

- Work-on, wire, connect/disconnect circuits, replace equipment and components **less than 50V.**

These additional tasks may only be performed **under the supervision of Blue cardholder.**

1. Replace electrical equipment with 50V and above, such as motors, fuses, transformers, starters, breakers, switches, etc.
2. **Before power is restored,** it is the responsibility of the Blue cardholder to inspect and insure that correct wiring and proper replacement parts have been used.
3. **The Green cardholder should save the failed components and keep them available for inspection by the Blue cardholder.**

Green Card

Under direct supervision of a Blue card holder, a Green cardholder can assist in making modifications to new/existing wiring or components. This practice will enable **Green** cardholder to enhance his/her skills in a supervised environment.

A Green Card certified associate may replace components on 600Volts or less as long as individual wires are not required to be disconnected and reconnected, for example components having modular plugs but not the wiring inside the modular plug. These tasks do not require Blue Card inspection before power is restored.

PLEASE NOTE: If there is any sign of melting, burning, or any heat related fault a Blue Card certified Person must inspect the circuit prior to restoring power.

The following list of equipment may be replaced by Green Card certified associate and the equipment power restored without Blue Card inspection. This is only allowed if the replacement part is identical (manufacturer, type, etc.) to the original part.

- Lamp Changes (plug or pin type)
- Ballast changes (plug-in type only)
- Servo Motors (with quick disconnects)
- Any 2 or 3 prong plug equipment covered under the single-source HECP which is known to be in good working order

PLEASE NOTE: If the replaced component fails soon after replacement, then a Blue Card certified person must inspect the circuit prior to restoring power

Blue Card

Electrical Work Allowed :

- For troubleshooting, inspecting and having access to electrical components where it is not possible to provide finger touch protection.

  Example of acceptable installation when this is applicable: Disconnects, Motors and Transformers.

- Can make modifications to existing wiring or components. May add new wiring or components to existing circuits, 600V or less

- Where non "like for like" electrical component replacement is required – Blue Badge inspection required if non-like for like:

  Make an assessment on the impact of an alternative component use.

  As appropriate, test must be performed to verify continued safe operation of the affected circuits.

Non- Finger Touch

* If Green Badge opens an energized cabinet and it is non-finger touch – close the cabinet, notify supervisor.
* Only where **NOT POSSIBLE** to provide "Finger Touch" protection,
  * a **Blue cardholder** or Contractor company journeyman or master electrician must perform the troubleshooting
  * Voltage rated gloves with leather protectors shall be worn as additional PPE.

BEFORE TESTING YOUR ELECTRICAL SAFETY KNOWLEDGE

REVIEW SGOP-0022 FOR FULL DETAILS OF ALL REQUIREMENTS OF VISTAKON'S ELECTRICAL SAFETY POLICY